

Figure 3: **CoT Behaviors in Synthetic Experiments:** (a) Each curve corresponds to a specific CoT strategy with fixed per-step difficulty. The color of the bar beneath each curve represents the optimal per-step difficulty (t) at each task difficulty. The progressively darker gradient colors indicates that harder tasks consistently favor higher per-step difficulty. (b) Change of the optimal CoT length with increasing model size across different task difficulty levels. As model size increases, the optimal CoT length decreases. For a fixed model size, harder tasks also exhibit longer optimal CoTs. (c) During RL training, the model policy gradually favors a shorter CoT that corresponds to the optimal length. length (e.g., by prompting with a control token indicating the desired number of operators t per step) or allow it to choose its preferred length. Further details are in Appendix E.

3.2 Scaling Laws of the Optimal CoT Length and Practical Insights

Our controlled experiments not only corroborate the CoT behaviors observed in real-world scenarios but also allow for a more fine-grained analysis. These findings uncover several key scaling behaviors of the optimal CoT length that shed light into the practical designs of LLM reasoning.

I. Harder-Tasks’ CoTs Peak at Longer Lengths (Adaptive CoT Length Matters): Our synthetic experiments further confirm the existence of an optimal CoT length, which manifests itself as an inverted U-shaped performance curve when plotting accuracy against the number of reasoning steps, as shown in Figure 1a. This clearly indicates that both "underthinking" (CoT too short) and "overthinking" (CoT too long) are detrimental, underscoring the critical benefit of generating CoTs with adaptive lengths tailored to the problem’s demands. Moreover, we observe that the optimal CoT length shifts right as the task difficulty T gets larger, indicating that solving a harder task optimally requires a longer CoT (also observable numerically from Figure 3b). This suggests that a good reasoning model should be able to vary CoT lengths w.r.t. the overall task complexity.

II. Harder Tasks Peak at Harder Sub-tasks (Adaptive Per-Step Computation Helps): Figure 3a illustrates how the number of operators per step (t) impacts model accuracy across varying task difficulties (T). The envelope curve, tracing peak performance, reveals that as tasks become more challenging (larger T), optimal performance is often achieved by CoTs that involve more complex computations *per step* (i.e., a larger optimal t^*). This suggests that for harder problems, simply increasing the number of simple steps may not be as effective as increasing the complexity of each sub-task the model tackles within the CoT. Current LLMs with fixed Transformer layers have limited intrinsic ability to adapt their per-step computational depth for different sub-tasks. This implies that their reasoning strategy might remain suboptimal. In contrast, recent advancements like looped Transformers, which enable adaptive recurrent depth [14, 8], could offer a more promising avenue for dynamically adjusting per-step computation to align with this observed need, potentially leading to better reasoning performance.

III. Stronger Models Achieve Optimal Performance with Shorter CoTs (Model-Aware CoT Data Matter): We also examine how model capability (number of layers) influences the optimal CoT length. Figure 3b indicates that, across different task complexities, the optimal number of CoT steps (N^*) consistently decreases as the model’s capability (number of layers) increases. This is because stronger models can effectively handle more complex sub-tasks in each step, thus requiring fewer overall steps to reach the solution optimally. This finding has significant implications for training data curation. It suggests that to achieve peak performance, models of different sizes or capabilities require CoT data tailored to their respective optimal per-step complexities. Current practices, such as using the same CoT datasets to train LLMs of varying sizes or directly distilling CoTs from large models to small ones without adapting complexity, may be suboptimal. For instance, a small model might struggle to learn effectively from overly complex CoT demonstrations designed for a larger

model. Our analysis advocates for training each model with CoT data of adaptive complexity, aligned with its specific capabilities, to help it reach its optimal reasoning performance.

IV. RL Training Converges to Optimal CoT Length (RL Calibrates Reasoning Behaviors):

As discussed in Section 2.2, RL training of LLMs leads to shorter CoT lengths. Our synthetic experiments further replicate this phenomenon. We take a GPT-2 model pre-trained on CoT solutions of equally mixed lengths for a task of difficulty $T = 24$ and apply RL using rule-based outcome rewards with PPO on VERL [30, 32]. Figure 3c shows the change of the sampled CoT lengths along RL: as training progresses, the model increasingly favors the CoT structure corresponding to the optimal length $N^* = 5$ that yields the peak accuracy (96%) on this task. This demonstrates that RL, by optimizing for task success, can implicitly guide the model’s CoT generation policy towards the optimal length regime, thereby exhibiting the *simplicity bias*. This offers a fresh perspective for understanding the benefits of RL in LLM training: even if the initial CoT data used for pre-training or supervised fine-tuning is suboptimal (e.g., misaligned with the model size or the task complexity), RL can help calibrate the model’s behavior towards generating more optimally-lengthy CoTs.

4 Theoretical Analysis: Why an Optimal CoT Length Exists

The empirical findings from both real-world and synthetic datasets consistently point to the existence of an optimal Chain-of-Thought (CoT) length. In this section, we provide a theoretical framework to explain this phenomenon, formalizing how factors like task decomposition and error accumulation interact to determine this optimal length, and how it scales with model capability and task difficulty. All proofs are deferred to Appendix G.

4.1 Theoretical Formulation

Akin to the arithmetic tasks we studied in Section 3, we use the following simple theory model to describe the CoT process.⁵ Let $N \in \mathbb{N}^+$ be the total number of steps in the CoT process. Let T denote the total number of operators in the given arithmetic task (a proxy for task difficulty). We assume that each CoT step consists of a sub-question q_i (e.g., $2 + 1 =$) and its answer as a_i (e.g., 3).

Definition 4.1 (CoT Process Probability). Given a task q with T total operators and a model θ , the probability of an N -step CoT that leads to a final answer a_{final} is:

$$P(a_{\text{final}}|q, \theta, N) = \prod_{i=1}^N \underbrace{P(q_i|H_{i-1}, q, \theta, N)}_{\text{sub-question}} \underbrace{P(a_i|q_i, H_{i-1}, q, \theta, N)}_{\text{sub-answer}},$$

where $H_k := [t_1, a_1, \dots, t_k, a_k]$ denotes the CoT history of the first k steps.

Let a_i^* denote the correct answer to subtask q_i and q_i^* denote the unique correct sub-question for simplicity. To estimate the final accuracy $A(N) = P(a_N = a_N^*|q, \theta)$, we need to estimate the sub-question accuracy $P(q_i = q_i^*|H_{i-1}, q, \theta)$ and the sub-answer accuracy $P(a_i = a_i^*|q_i, H_{i-1}, q, \theta)$.

For the **sub-question accuracy**, following experimental observation (in Appendix E.2), we assume that the error rate of generating each question q_i , denoted by $\sigma(T) \in [0, 1]$, is positively correlated with the total number of operators T . Intuitively, as the number of operators increases, extracting the correct subtask becomes more challenging. For the **sub-answer accuracy**, it is clear that when given subtask q_i , $P(a_i = a_i^*|q_i, H_{i-1}, q, \theta)$ is independent of the history reasoning steps H_{i-1} and is only influenced by the model θ and the difficulty of the subtask q_i . For each model, we define its capability M based on the reasoning boundary [5], e.g., the maximum number of operators the model can directly solve per step; thus, a stronger model has a larger M . We define the error rate of each subtask answer as $E(N, M, T) \in [0, 1]$.

Proposition 4.2. *The total accuracy of N -step reasoning is*

$$A(N) = P(a_{\text{final}} = a_{\text{final}}^*|q, \theta, N) = \alpha ((1 - E(N, M, T))(1 - \sigma(T)))^N, \quad (1)$$

where α denotes a constant value independent of N .

⁵Note that we do not explicitly model various cognitive reasoning behaviors (reflection, verification, backtracking) but instead regard them as one of the many ways that one can decompose a task into subtasks to ease problem solving, which can be understood as a part of our task decomposition formulation in a general sense.

Proposition 4.2 establishes the quantitative relationship between CoT’s final performance A and reasoning length N . Once we obtain estimates for $E(N, M, T)$ and $\sigma(T)$, we can determine the optimal CoT length N^* as a function of the model capability M and task complexity T .

Simple Case with Linear Error. To gain intuition, we first consider a simple case where the sub-question error scales linearly with T , i.e., $\sigma(T) = T/C$, where C is a constant representing the maximum task difficulty models are trained to handle. Throughout this analysis, we assume $\sigma(T) \leq 0.9$ to restrict our discussion to tasks that are within the model’s training regime. Otherwise, it would be unreasonable to claim the model has learned to solve such problems. We also assume the sub-answer error rate scales linearly with harder tasks, fewer steps, and weaker models as $E(N, M, T) = (T/N)/M = T/(NM)$. Under these simplified conditions, we can derive the following closed-form expression for the optimal CoT length.

Theorem 4.3 (Optimal CoT Length). *For a given model capability M and task difficulty T , the total accuracy $A(N) = \alpha[(1 - T/C) \cdot (1 - T/(NM))]^N$ (Eq. (1)) initially increases and then decreases as N increases (forming an inverted U-shape). Thus, there exists an optimal CoT length:*

$$N^*(M, T) = \frac{TZ}{M(Z + 1)}, \quad (2)$$

that maximizes $A(N)$, where $Z = W_{-1}(-1 - T/(Ce))$, and $W_{-1}(x)$ is the smaller real branch of the Lambert W function satisfying $we^w = x$, and e is the natural number.

This theorem formally establishes the inverted U-shaped curve and provides an explicit form for N^* . From this, we can formally prove the first three scaling behaviors characterized in Section 3.2.

Corollary 4.4 (Scaling laws of Optimal CoT Length). *Based on Theorem 4.3, one can derive:*

- $N^*(M, T)$ increases monotonically with T , i.e., harder tasks require more reasoning steps to attain the optimal performance.
- The optimal number of operators per step $t^* = T/N^*(M, T) = M(1 + 1/Z)$ increases monotonically with T . This aligns with the envelope curve result (Figure 3a).
- $N^*(M, T)$ decreases monotonically with M , i.e., more capable models require fewer reasoning steps to attain the optimal performance, reflecting the simplicity bias.

Extension to Broader Scenarios. Here, we adopt a simple linear model to facilitate intuitive understanding. However, this analysis can be extended to more general settings, including general error functions (only with mild assumptions of monotonicity and convexity) and stochastic error models, where each subtask may exhibit a different error rate. These extensions introduce additional technical subtleties but follow the same underlying principles. We defer this part to Appendix F.

4.2 Why does RL Exhibit Simplicity Bias?

The analysis above also provides a natural understanding of RL’s simplicity bias (Section 3.2). As in the arithmetic task, we generate samples within a finite discrete action space $\mathcal{A} = \{N_1, N_2, \dots, N_k\}$ during RL that receive binary outcome rewards. This reduces to a stateless bandit: each N_i yields reward $r \in \{0, 1\}$ with probability $A(N_i)$ (from Proposition 4.2). Let us parameterize a softmax policy $\pi_\theta(N_i) = \frac{e^{\theta_i}}{\sum_j e^{\theta_j}}$, and define the RL objective as $J(\theta) = \sum_{i=1}^k \pi_\theta(N_i) A(N_i)$. As a result, the policy-gradient becomes $\nabla_{\theta_i} J = \sum_{j=1}^k A(N_j) \pi_\theta(N_j) (\delta_{ij} - \pi_\theta(N_i))$.

Corollary 4.5 (RL Converges to Optimal CoT Length). *For gradient ascent on $J(\theta)$ with sufficiently small step size, the policy converges to a deterministic policy $\pi_\theta(N_i) = 1$ iff $i = \arg \max_j A(N_j)$. Thus, RL training converges to the optimal CoT length $N^* = \arg \max_{N \in \mathcal{A}} A(N)$.*

This corollary shows that RL will automatically discover the optimal length (usually shorter length) through optimizing the reward function and exhibit a decreasing CoT length as in the simplicity bias phenomenon. In this way, our theory offers an explanation of the optimal CoT length, its scaling behavior and RL’s simplicity bias within a unified framework.

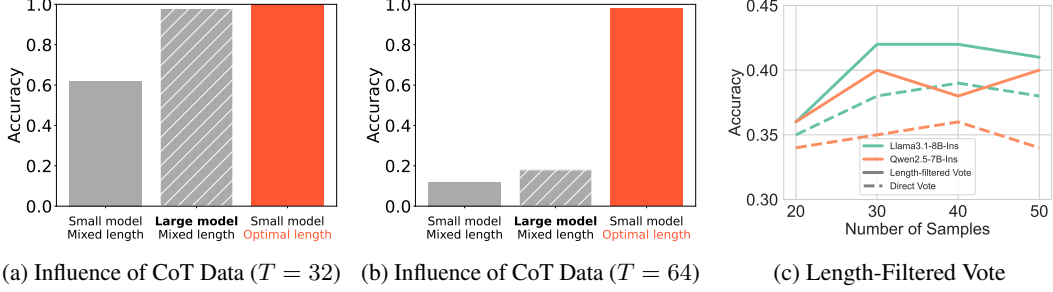


Figure 4: (a) and (b) compare model performance under different pretraining data distributions: Mixed Length (uniform over all lengths) vs. Optimal Length (only optimal-length solutions). Despite its smaller size, the small (6 layer) model trained on optimal-length data outperforms the large (9 layer) model trained on mixed-length data, with the performance gap widening as task difficulty increases. (c) Our Length-Filtered Vote method consistently outperforms vanilla majority vote on the GPQA dataset, maintaining strong performance even as the number of samples increases.

5 Practical Applications of Optimal CoT Length

Guided by the understanding above, in this section, we illustrate via some proof-of-concept experiments that adapting LLM training and inference configurations to the optimal CoT length can improve the model’s reasoning performance.

5.1 Training with Data of Optimal CoT Length

Training with Optimal-Length CoT Data: The existence of an adaptive, optimal CoT length suggests that one should design the CoT training data adaptively to fully optimize the model’s reasoning performance. To examine the influence of the CoT length of the training data, we train a model on a specialized dataset that contains CoT solutions with lengths known to be optimal for the given model size and task difficulty (T). We compare this model against a baseline model trained on a dataset of CoT solutions with uniformly distributed step lengths t . During testing, models were allowed to freely choose their CoT strategy.

Results. As shown in Figures 4a and 4b, the model trained on optimal-length CoTs significantly outperforms the models trained on mixed-length solutions. Remarkably, a smaller model (e.g., 6 layers) trained on optimal-length data can even outperform a larger model (e.g., 9 layers) trained on randomly chosen CoT lengths. This proof-of-concept experiment underscores the critical influence of the suitability of the CoT length in training data for the model. While it is generally hard to exactly estimate optimal CoT lengths in real-world problems, our theoretical and empirical studies provide valuable guidelines for a coarse estimate. We leave more in-depth studies to future work.

5.2 Adaptive Length-Filtered Vote at Inference Time

The observation that CoTs of optimal length yield higher accuracy suggests that inference-time strategies could benefit from this insight. Standard approaches like majority voting over multiple sampled CoTs, such as self-consistency [37], treat all valid reasoning paths equally, regardless of their length. However, paths that are too short (underthinking) or too long (overthinking and error-prone) may contribute noisy or incorrect answers to the voting pool.

Inspired by our findings, we propose **Length-Filtered Vote**, an adaptive method that enhances standard majority voting by preferentially weighting or exclusively considering answers derived from CoTs whose lengths fall within a proper range. Specifically, in majority vote, given a model f_θ , a question q , a ground truth answer a^* , we first sample a set of answer candidates $c_1, \dots, c_n \stackrel{i.i.d.}{\sim} f_\theta(q)$ independently. After that, instead of a direct vote, we group the answers by their corresponding CoT length $\ell(c_i)$ (discussed in Appendix C) into groups with equal bin size D (by default, we set $D = 2$), denoted as $\{L_j\}_{j=1}^m$. As our theory suggests that the prediction accuracy is peaked around a certain range of CoT length, we identify such groups through the prediction uncertainty of the answers within each group, based on the intuition that lower uncertainty implies better predictions. Specifically, we